E-commerce Customer Segmentation & Revenue Analysis

1. Introduction

Understanding customer behavior is critical for improving revenue, customer retention, and business growth in e-commerce. This project focuses on analyzing transactional data to identify purchasing patterns and segment customers based on their behavior. The insights derived from this analysis can help businesses make data-driven decisions.

2. Project Objectives

- Understand customer purchasing behavior
- Identify revenue drivers
- Perform customer segmentation using RFM analysis
- Apply clustering techniques to group customers
- Generate business recommendations

3. Dataset Description

The dataset contains transactional data from an e-commerce business with the following features:

- InvoiceNo – Transaction ID
- StockCode – Product ID
- Description – Product description
- Quantity – Number of items purchased
- InvoiceDate – Transaction date
- UnitPrice – Price per item
- CustomerID – Unique customer ID
- Country – Customer location

4. Project Workflow

1. Data Loading and Understanding
2. Data Cleaning and Preprocessing
3. Exploratory Data Analysis
4. Customer Behavior Analysis
5. RFM Segmentation
6. Customer Clustering
7. Insights and Business Recommendations

5. Data Cleaning and Preprocessing

Data cleaning was performed to ensure data quality before analysis. Missing values in CustomerID were removed, cancelled transactions were filtered out, and duplicate entries were eliminated. A new revenue column was created using Quantity and UnitPrice. InvoiceDate was converted into datetime format for time-based analysis.

6. Exploratory Data Analysis

Exploratory Data Analysis was conducted to understand revenue patterns, top products, and geographic distribution. The analysis revealed that revenue was highly skewed, with a small number of customers contributing significantly.

6.1 Revenue Trend

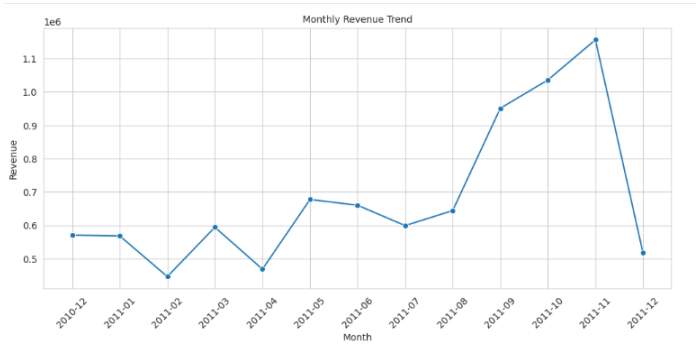


Fig 1: Monthly Revenue Trend

6.2 Top Products

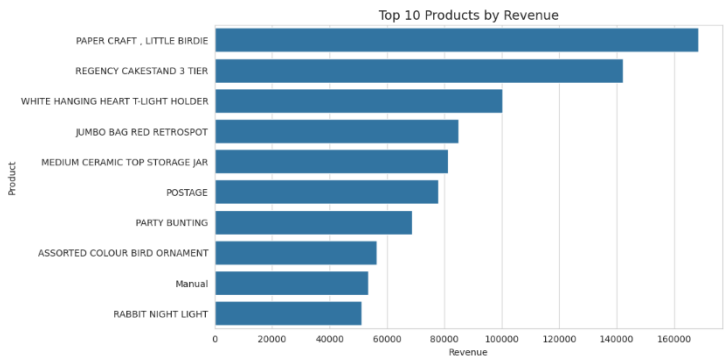


Fig 2: Top Products by Revenue

6.3 Country Analysis

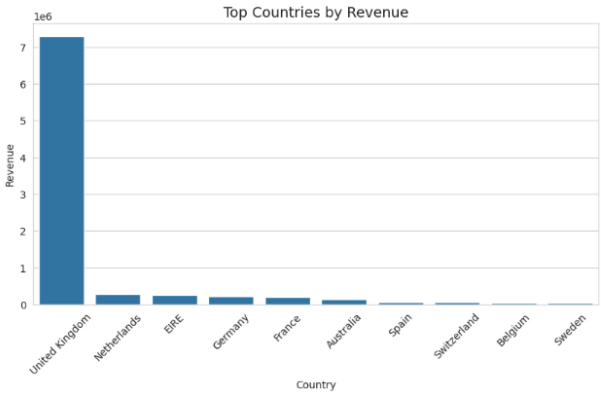


Fig 3: Revenue by Country

Fig 4: Revenue by country excluding UK

7. Customer Behaviour Analysis

Customer behaviour analysis was performed to understand purchasing patterns and customer engagement levels. This helps identify opportunities to improve retention and increase customer value.

7.1 Purchase Frequency Distribution

Customer purchase frequency was analyzed to determine how often customers make purchases.

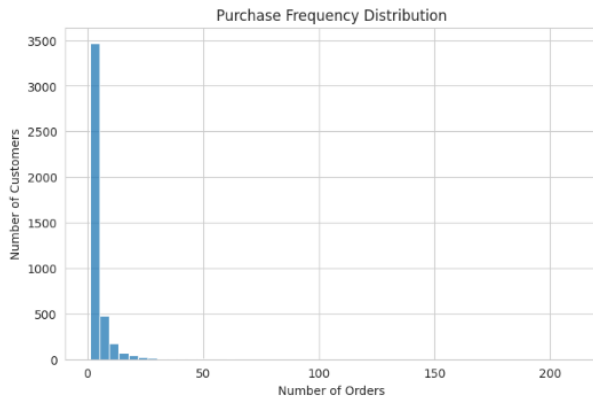
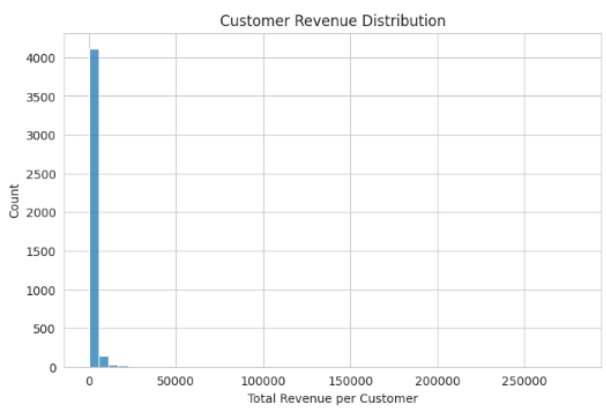


Fig 5. Purchase Frequency Distribution

7.2 Revenue per Customer Distribution

Revenue generated by each customer was analyzed to understand spending behaviour. Since the distribution was highly skewed, log transformation was applied for better visualization.



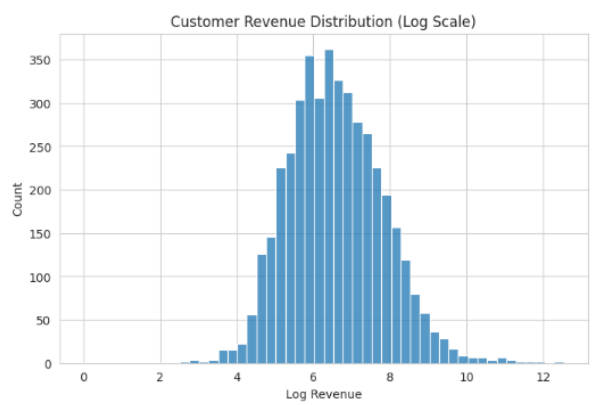


Fig 6. Revenue per Customer Distribution

Fig 7. Revenue per Customer Distribution (Log Scale)

8. RFM Analysis

Customers were segmented using Recency, Frequency, and Monetary metrics. This helped identify high-value customers and customers at risk of churn.

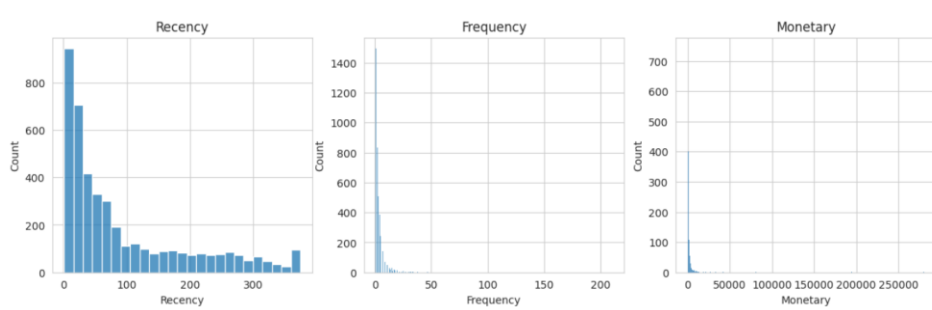


Fig 8. RFM distribution

9. Customer Clustering

K-Means clustering was applied to group customers into distinct segments including VIP, Loyal, Moderate, and At Risk customers.

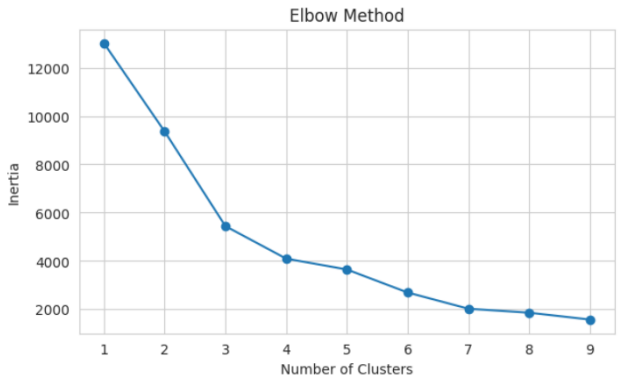


Fig 9. Elbow Method

Fig 10. Cluster Visualization

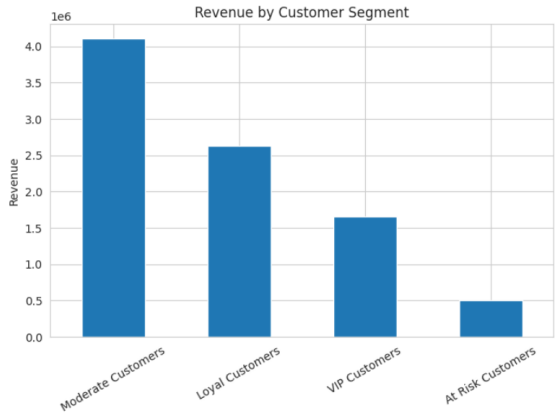


Fig 11. Cluster Visualization

10. Key Results

- Moderate customers contributed ~46% of revenue
- Loyal customers contributed ~29.5% of revenue
- VIP customers contributed ~18.6% of revenue
- At-risk customers contributed ~5.7% of revenue

11. Business Recommendations

- Retain VIP customers
- Upsell moderate customers
- Engage loyal customers
- Re-engage at-risk customers

12. Conclusion

This project demonstrated how customer segmentation and clustering can help identify valuable customer groups and improve business strategy. The insights can be used to improve retention and drive revenue growth.